

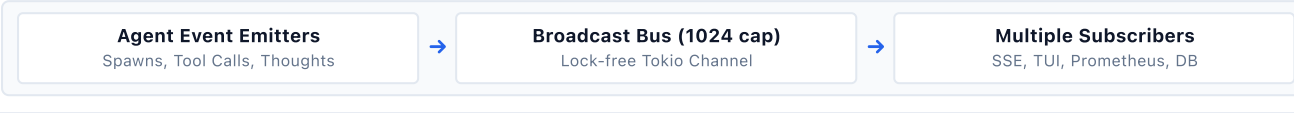
The Broadcast Message Bus

Real-Time Event Distribution Across Swarms, UI Dashboards & External Channels

Decoupled Asynchronous Telemetry Architecture

Communication across sub-agents, persistence layers, and client dashboards occurs over a centralized, high-throughput **Tokio broadcast message bus** (`event_tx`).

Message Bus Architecture & Event Flow



1. SERVER-SENT EVENTS (SSE)

Streams live agent events to web and desktop dashboards, updating UI progress bars and sparklines in real time.

2. PROMETHEUS SCRAPERS

The metrics middleware consumes event counters, exporting real-time tool durations and token velocity to Grafana.

3. SQLITE AUDIT LEDGER

Records immutable execution logs, tool inputs, and decision outcomes for permanent compliance replay.

High-Throughput Guarantee: The lock-free broadcast channel handles over 50,000 inter-agent events per second with sub-microsecond latency.